

Non-destructive testing of concrete structures

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Types on NDT

- tests for assessing the strength (this session)
- tests for assessing the durability (next session)
- tests for measuring the integrity (third session)

ACI 228.2R-13

**Report on Nondestructive Test
Methods for Evaluation of Concrete
in Structures**

Reported by ACI Committee 228



American Concrete Institute®

**Report on Methods
for Estimating In-Place
Concrete Strength**

Reported by ACI Committee 228

ACI 228.1R-19

Introduction

- Routine tests is difficult, time consuming and expensive
- No method for assessment of location, size of reinforcement and its corrosion, concrete cover thickness to steel in existing structures
- Visual inspection is subjective and for surface only
- Number of NDT equipment are now available.
- NDTs are gradually becoming acceptable method to evaluate in-service performance of structures.

Advantages of NDT

- Portable, handy, battery operated and easy to use available in different capacities,
- Less time consuming and more realistic as tests are conducted on actual structures.
- Test can be performed at several locations at various ages of concrete.
- Confirmation of locations of suspected distress or deterioration

Limitations of NDT

- The equipment and technique are relatively new. Many of them are yet to be standardized and accepted.
- Most of the equipment are not available locally.
- Much experience and expertise is required for analysis and proper interpretation of data to get satisfactory results.
- It is desirable to develop own calibration curves for different equipment for getting good results.
- It is desirable to use combination of two or more equipment for getting reliable results.

Tests for assessing strength

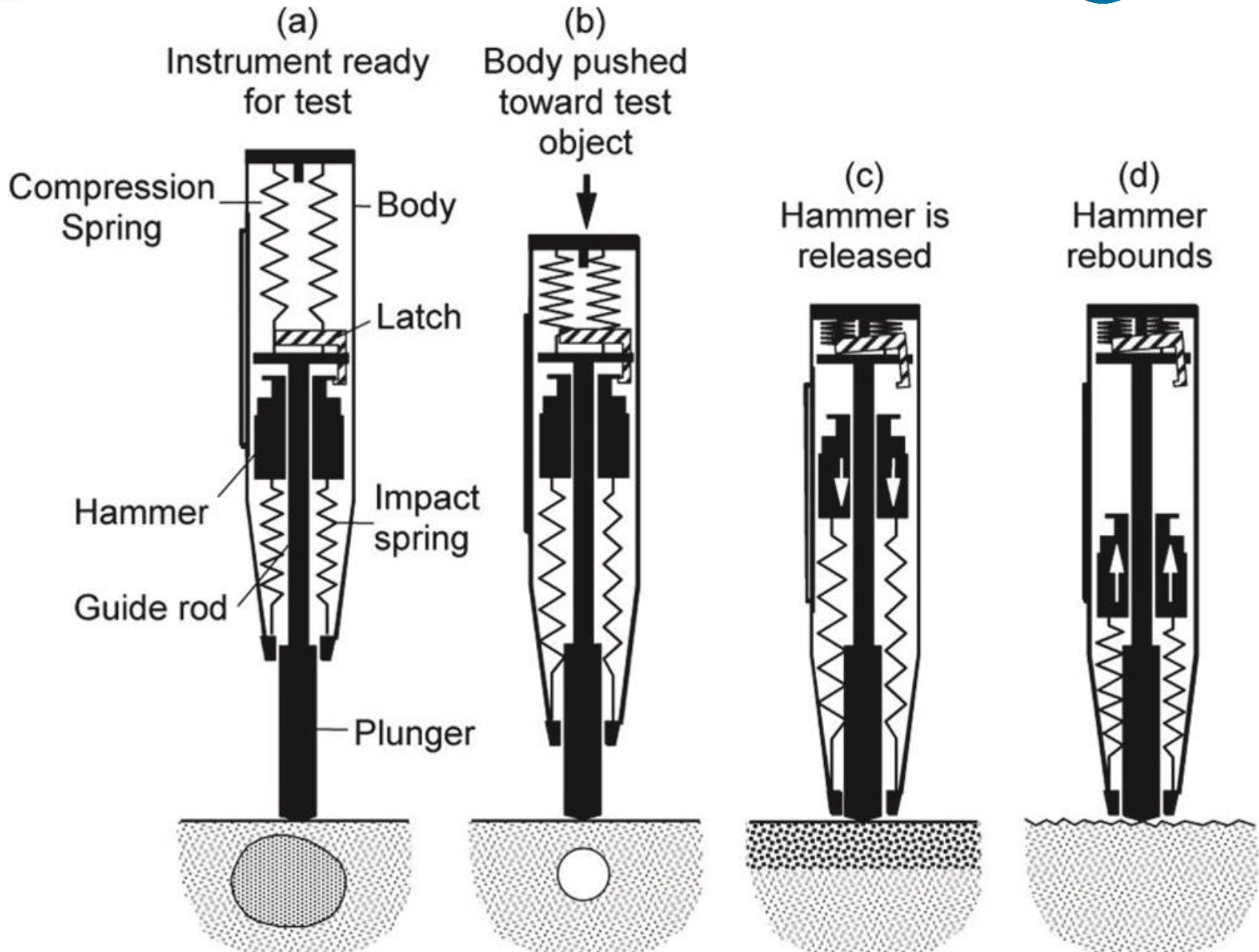
Test method	Intended purpose	Classification	Reference
Rebound hammer test	Surface hardness of concrete	NDT	BS EN 12504-2: 2001; ASTM C805
Ultrasonic pulse velocity	Non-uniformity and density of concrete	NDT	BS EN 12504-4: 2004; ASTM C597
Pull-out tests: Capo and Lok test	Pull-out strength of concrete cover	PDT	BS EN 12504-3: 2005; ASTM C900; Bungey and Soutsos, 2001
Pull-off test	Pull-off surface tensile strength of concrete	PDT	BS EN 1542: 1999; ASTM C1583; Bungey and Soutsos, 2001
Windsor probe penetration resistance test	Probe penetration resistance of concrete cover	PDT	BS 1881: Part 207; Bungey and Soutsos, 2001
Break-off test	Break-off strength of concrete cover	PDT	BS 1881: Part 207; Bungey and Soutsos, 2001
Maturity measurement	Strength development	NDT	ASTM C1074-02; Malhotra and Carino, 2004

Rebound Hammer

- It is handy, portable mechanical equipment not requiring any consumable
- It consists of a spring controlled mass that slides on a plunger within a tubular housing.
- When the plunger is pressed against the surface of the concrete, the spring controlled mass rebounds. The extent of such rebound depends upon the surface hardness of the concrete and is taken to be related to the compressive strength of the concrete.

Rebound Hammer

- Based on the principle that the rebound of an elastic mass depends on the hardness of the surface against which the mass impinges.
- It provides a convenient rapid indication of the compressive strength of concrete .
- The surface to be tested should be dry, smooth and clean
- The point of impact should be at least 20 mm away from any edge or shape discontinuity
- Should be held at right angles to the surface of the concrete member



Limitations

- Gives the strength of concrete at surface only
- Internal micro cracking, flaws etc. cannot be detected.
- Probable accuracy of prediction of concrete strength is $\pm 25\%$
- On carbonated concrete surface results will not be accurate

Factors affecting rebound value

- Wetness of the surface
- Surface finish of the concrete
- Inclination of the rebound hammer
- Type of aggregate
- Age of concrete

Ultrasonic pulse velocity meter

- The velocity of Ultra sonic pulse in a material depends on its density and elastic properties, which in turn are related to the quality and the strength of the material. Comparatively higher values is obtained when concrete quality is good.

Pulse velocity by cross probing(km./sec.)	Conc. quality grading
Above 4.5	excellent
3.5 to 4.5	Good
3.0 to 3.5	Medium
Below 3.0	doubtful

Ultrasonic pulse velocity standards

• Concrete:

- BS 12504-4
- BS 1881-203
- ASTM C597
- ISO 1920-7

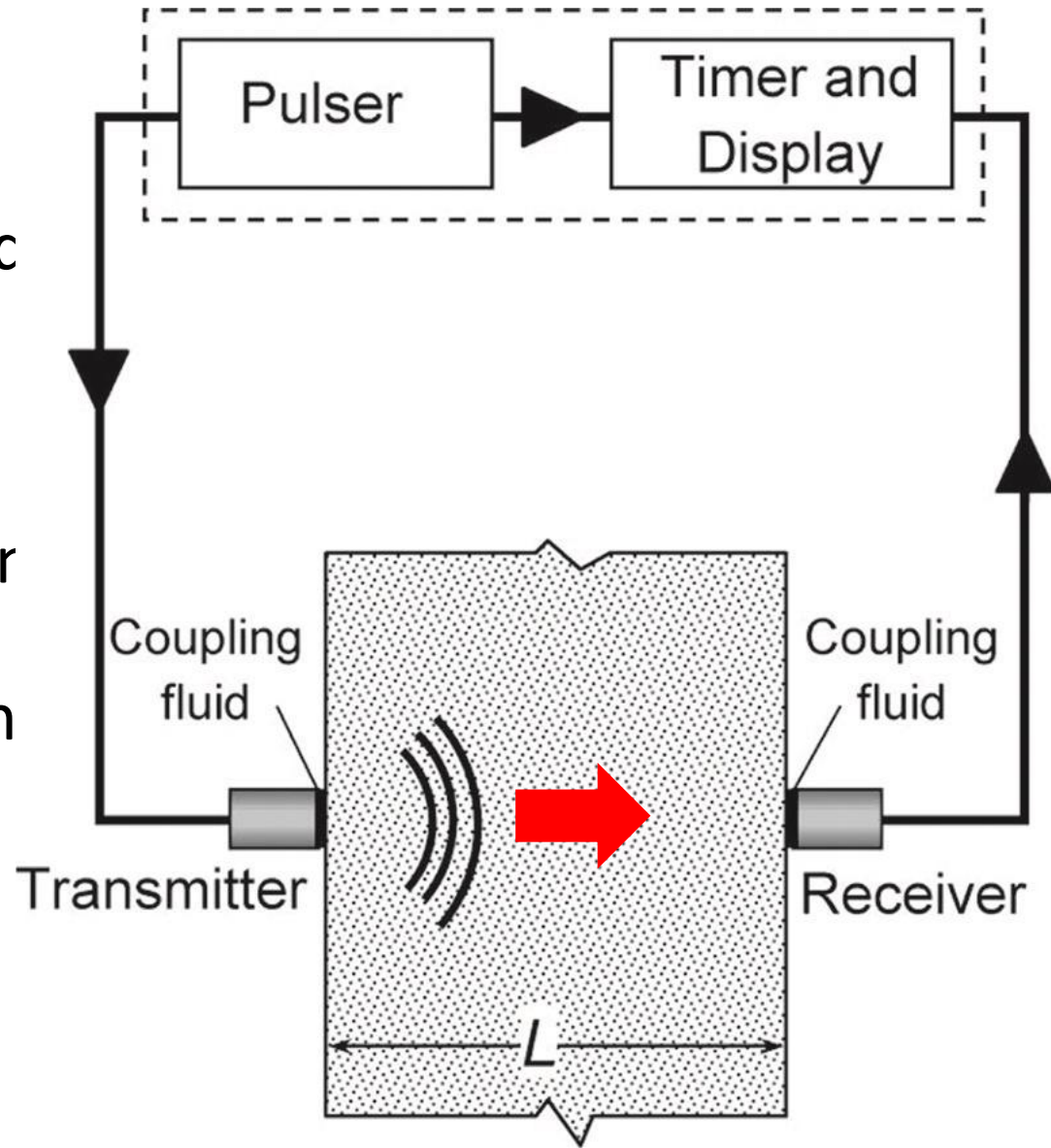
• Rock:

- ASTM D2845

Country	Designation	Year
Belgium	NBN 15-229 1976	1976
Brazil	ABNT 18:04.08.001 1983	1983
Bulgaria	BDS 15013-80 1980	1980
Czech Republic	CSN 731371 1981	1981
Denmark	DS 423.33 1984	1984
Germany	Draft same as ISO/DIS 8047 1983	1983
Hungary	MI 07-3318 1994	1994
International	ISO/DIS 8047 1983	1983
Mexico	NOM-C-275-1986 1986	1986
Poland	PN-B-06261	1974
RILEM	NDT 1 1972	1972
Rumania	C-26-72 1972	1972
Russia	GOST 17624 1987	1987
Scandinavia	NT BUILD 213 1984	1984
Spain	UNE 83-308-86 1986	1986
Sweden	SS 137240 1983	1983
United Kingdom	BS 1881: Part 203 1986	1986
USA	ASTM C 597 1983	1983
Venezuela	COVENIN 1681-80 1980	1980
Yugoslavia	JUS U.M1.042 1982	1982

Apparatus to measure UPV

- Ultrasonic pulse is produced and transmitted by a transducer held in contact with concrete surface, which travels through a known length in concrete and is received by a second transducer held in contact with another or same surface of concrete member.

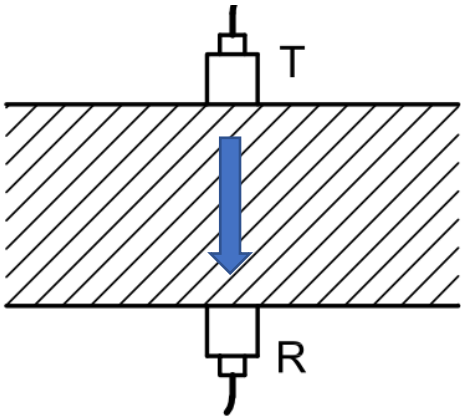




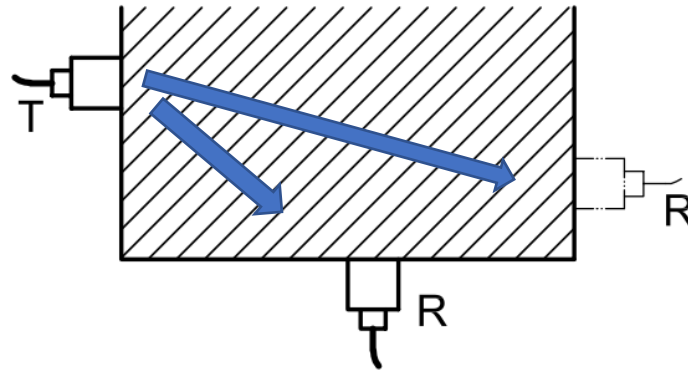
Benefits

- Speed
- ease of carrying out

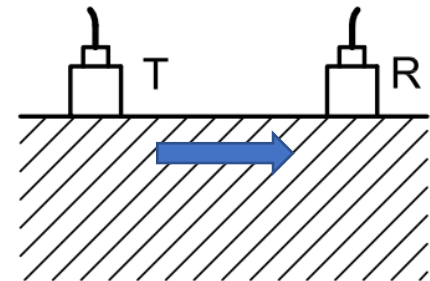
Methods of propagating and receiving ultrasonic pulses



a) Direct transmission

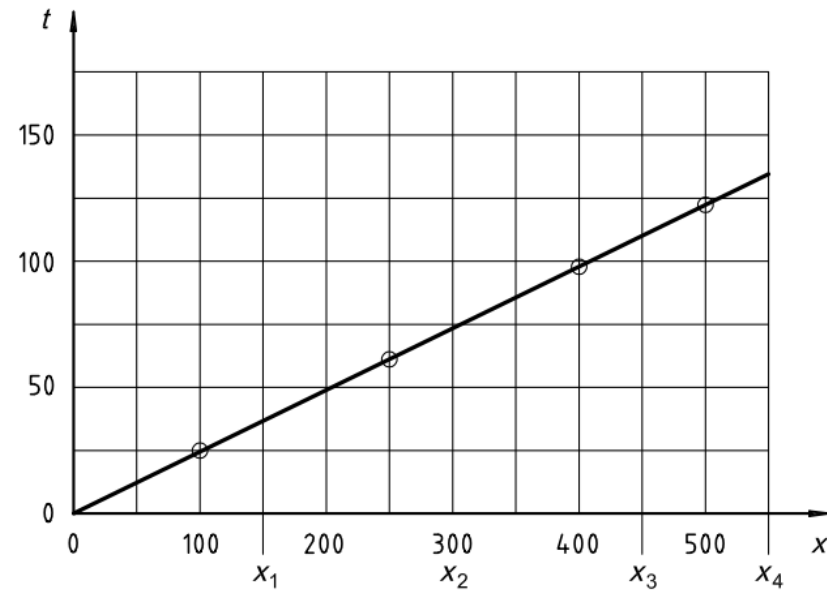
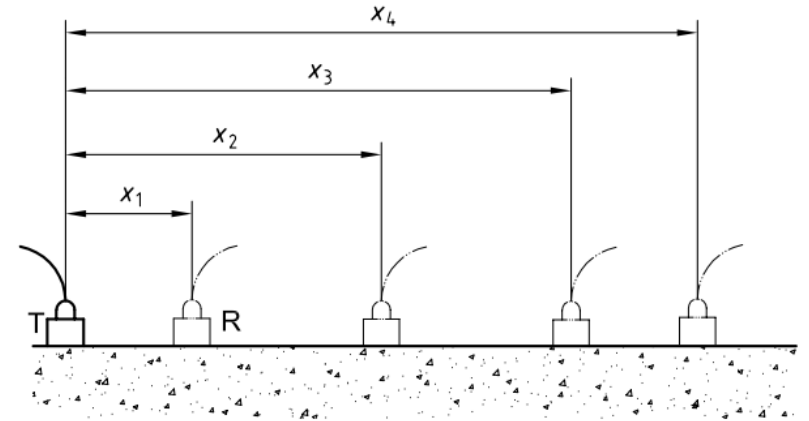
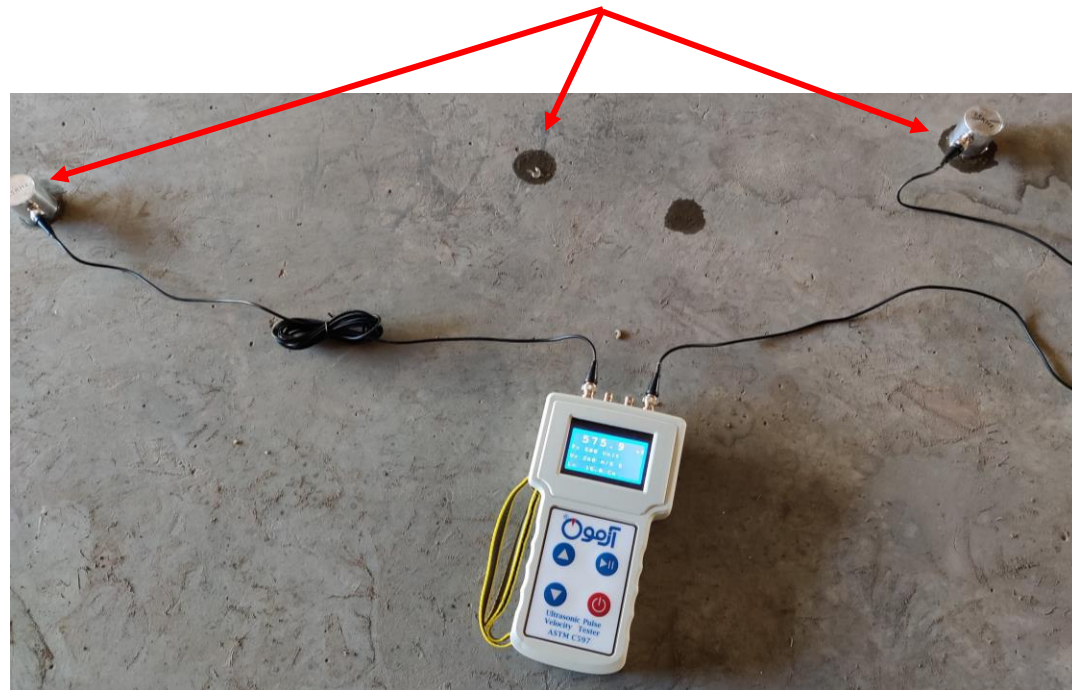


b) Semi-direct transmission



c) Indirect transmission

Pulse velocity determination by indirect transmission



Key

x is the distance, in millimetres
 t is the transit time, in microseconds

T is the transmitter
 R is the receiver



L=18 cm

V=324 m/s



Passage time	from 0.1 to 2000 microseconds with an accuracy of 0.1 microseconds
Pulse rate	1, 2, 5, 10 per second, selectable
Maximum output voltage	500 volts
Frequency range	from 24 to 150 kHz
PC connection	Yes - with micro usb output
Oscilloscope connection	Yes
Rechargeable battery capacity	2800 MAh

AZMOON GROUP
STOP

OPTION
L= 10.5 Cm

OPTION
250 Volt

. 0 uS
P= 150 Volt
U= 0 m/S
L= 10.5 Cm

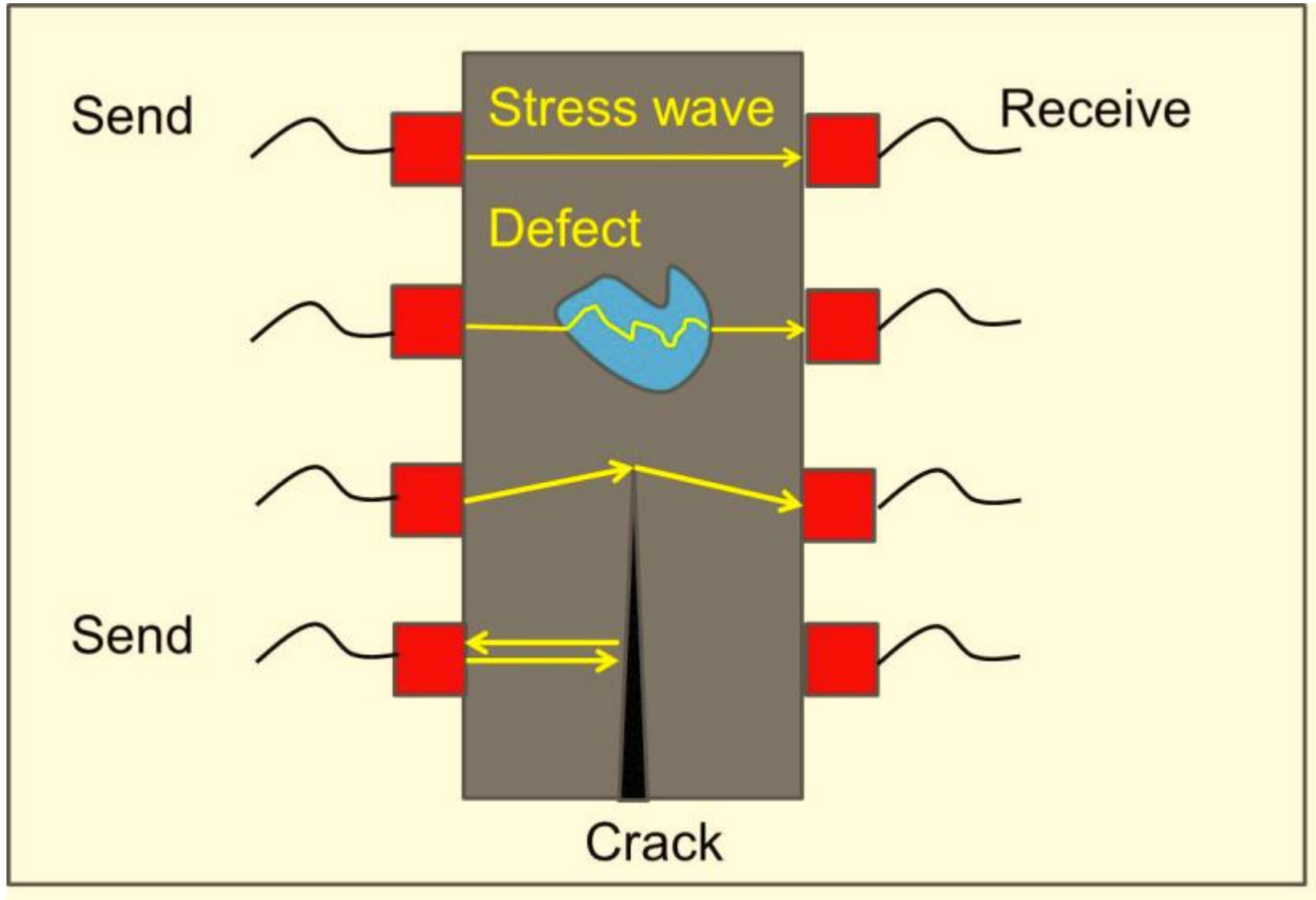
OPTION
Calibration
. 0 uS
125

54.34 Kh
9.2 uS

Applications of UPV

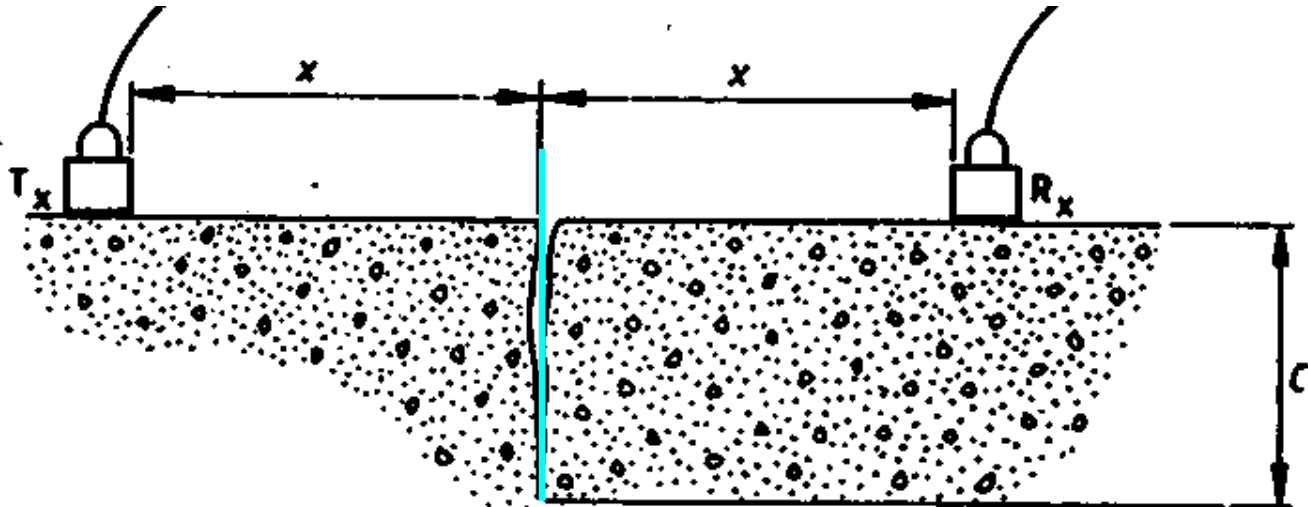
- Quality Control
- layer thickness
- Crack depth
- Ed-modulus
- Compressive strength

Quality



Crack Depth

$$C = x * \sqrt{\frac{(4t_1^2 - t_2^2)}{(t_2^2 - t_1^2)}}$$



Ed-modulus

The method is more effectively used for comparison amongst different part of the structures

The dynamic Young's Modulus of elasticity

$$E_d = \rho v^2 \frac{(1 + \nu) (1 - 2\nu)}{(1 - \nu)}$$

ρ = density in kg./m.³

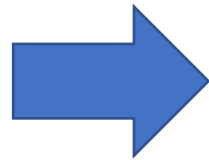
v = velocity in m/sec.

ν = dynamic Poisson's ratio

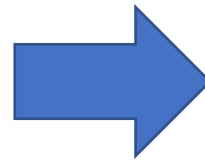
E-modulus

$$M = \rho V_b^2$$

$$G = \rho V_s^2$$



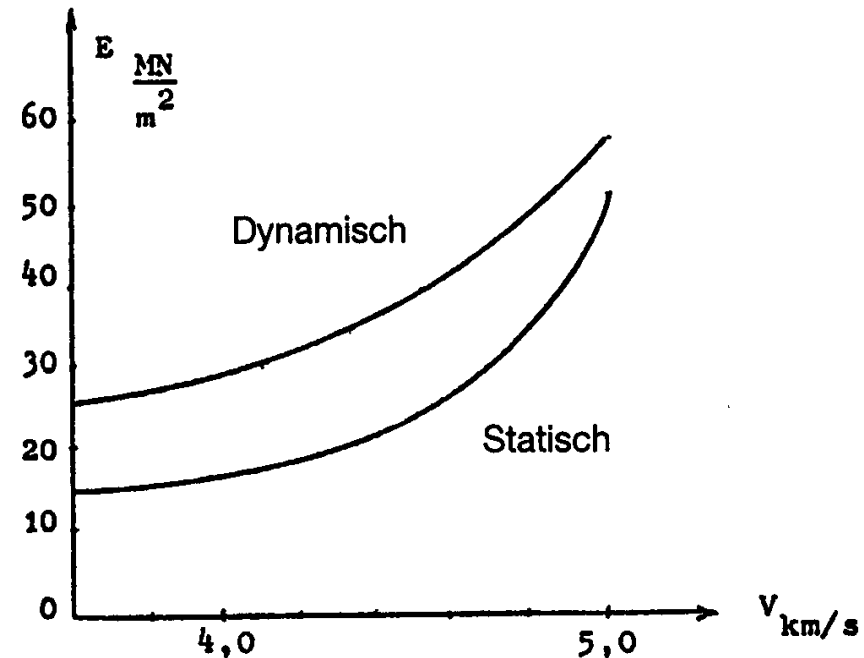
$$\nu = \frac{M - 2G}{2M - 2G}$$



$$E = 2G(1 + \nu)$$

Empirical relationship between static and dynamic moduli of elasticity and pulse velocity

Pulse velocity	Modulus of elasticity	
	Dynamic	Static
km/s	MN/m ²	MN/m ²
3.6	24 000	13 000
3.8	26 000	15 000
4.0	29 000	18 000
4.2	32 000	22 000
4.4	36 000	27 000
4.6	42 000	34 000
4.8	49 000	43 000
5.0	58 000	52 000



Compressive strength

$$f_c = a \times V^b$$

- type of aggregate
- type of cement
- water/cement ratio
- mix proportions
 - age of concrete
 - curing conditions
 - local moisture content

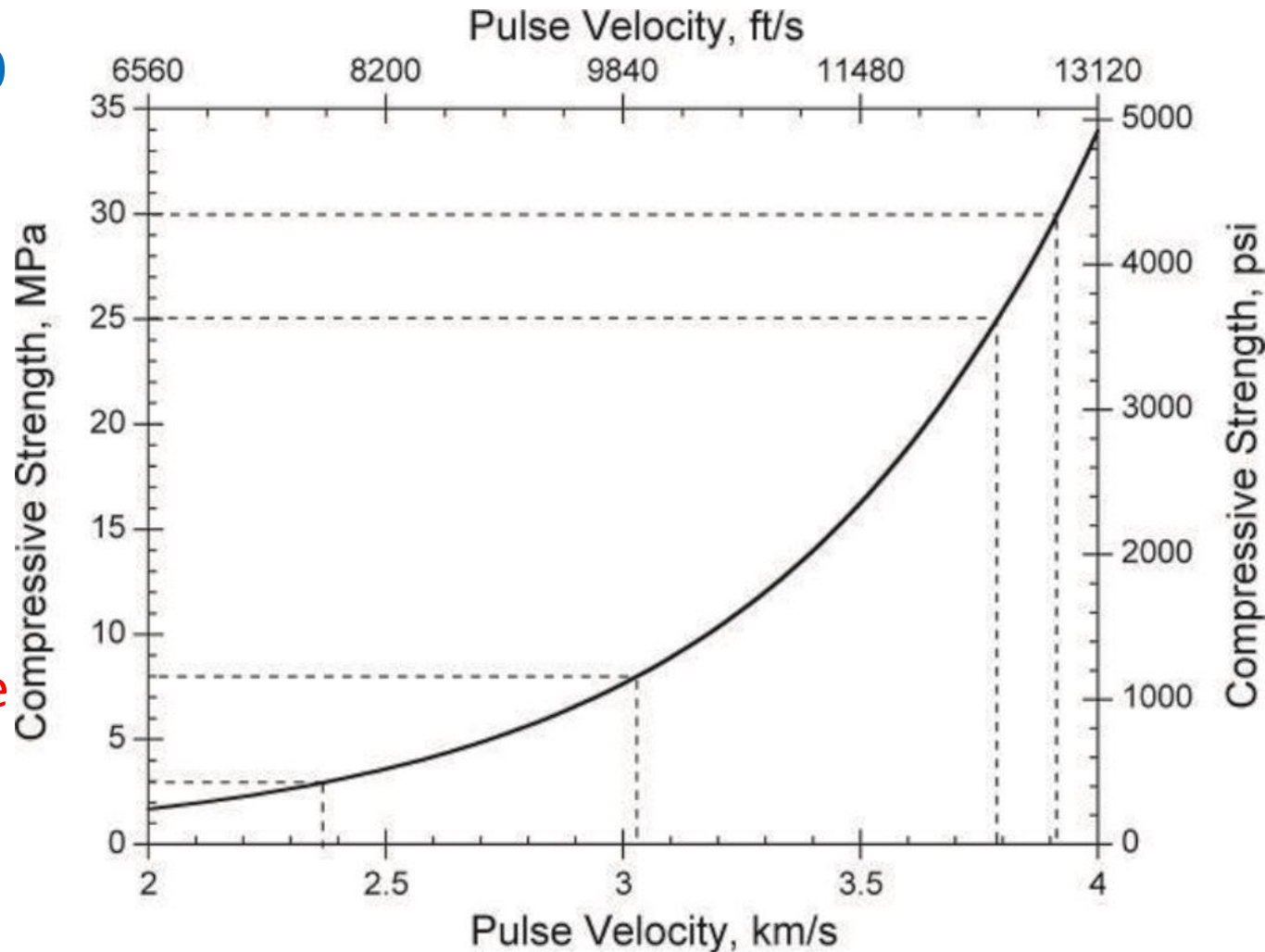


Table 2: Concrete mix proportion for verification of UPV-strength relationship curves

w/c	Cement/kg	Sand/kg	C.A (10 mm)/kg	Water/L
0.35	15.4	30.9	61.7	4.6
0.40	15.4	30.9	61.7	6.2
0.50	15.4	30.9	61.7	7.7
0.60	15.4	30.9	61.7	9.2
0.75	15.4	30.9	61.7	11.6

Corpus ID: 138032883

Table 3: Comparison results of UPV-strength relationship

w/c	Av.velocity/ms	Theoretical C.S/MPa	Experimental C.S/MPa
0.35	4755.00	33.15	32.25
0.40	7505.60	40.30	41.00
0.50	6305.00	29.00	30.40
0.60	5446.60	22.50	21.40
0.75	4529.20	19.00	20.65

FACTORS AFFECTING PULSE VELOCITY

- Moisture content
- Degree of coupling
- Reinforcement
- Mix proportion
- Age of concrete
- Stress level in concrete
- Concrete temperature

Dimension Effects

- The natural frequency of transducer should be between 20 to 150 kHz
- High frequency transducer for short path lengths and low frequency transducers for long path length
- Provision of adequate acoustical coupling between transducer and concrete surface
- Min. path length 100 mm. If max. size of aggregate is 20 mm
- Min. path length 150 mm. If max. size of aggregate is bet. 20 to 40 mm

Lateral dimensions Effects

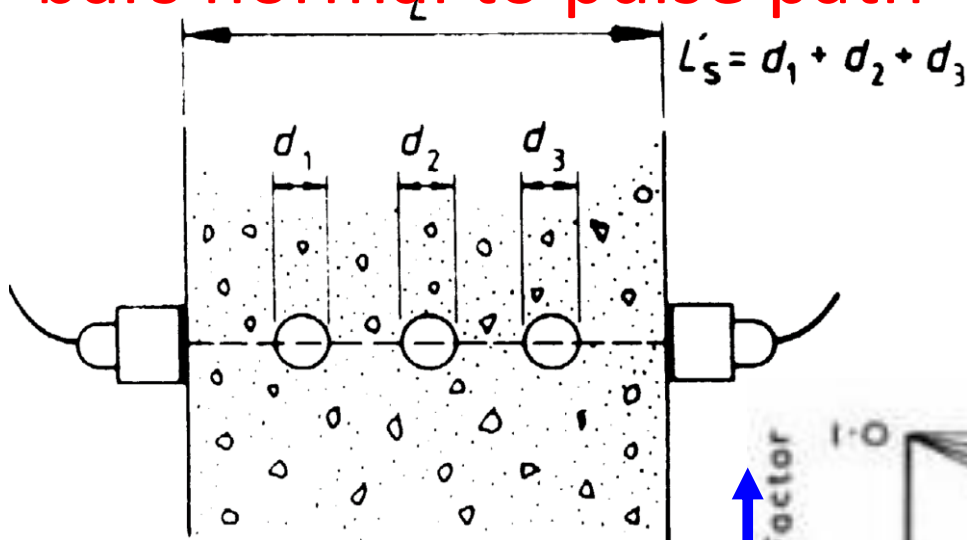
Transducer frequency kHz	Pulse velocity in concrete		
	3,50 km/s	4,00 km/s	4,50 km/s
	Minimum permissible lateral specimen dimension, in mm		
24	146	167	188
54	65	74	83
82	43	49	55
150	23	27	30

Higher sound velocity requires
larger dimensions

Higher Frequency allows
smaller dimensions

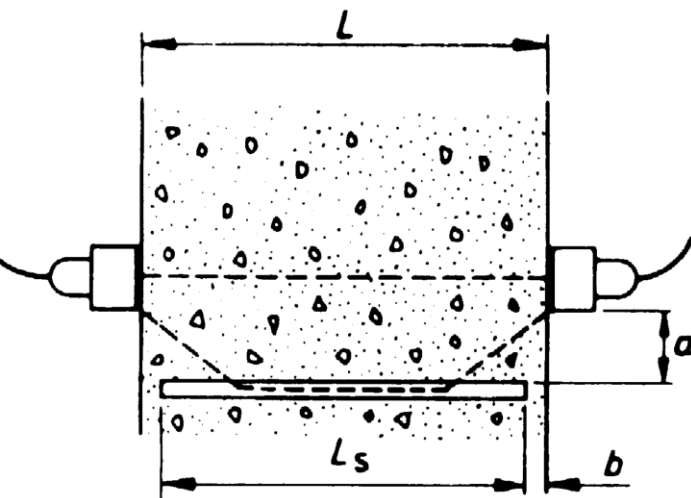
Influence of steel reinforcement on pulse velocity

bars normal to pulse path

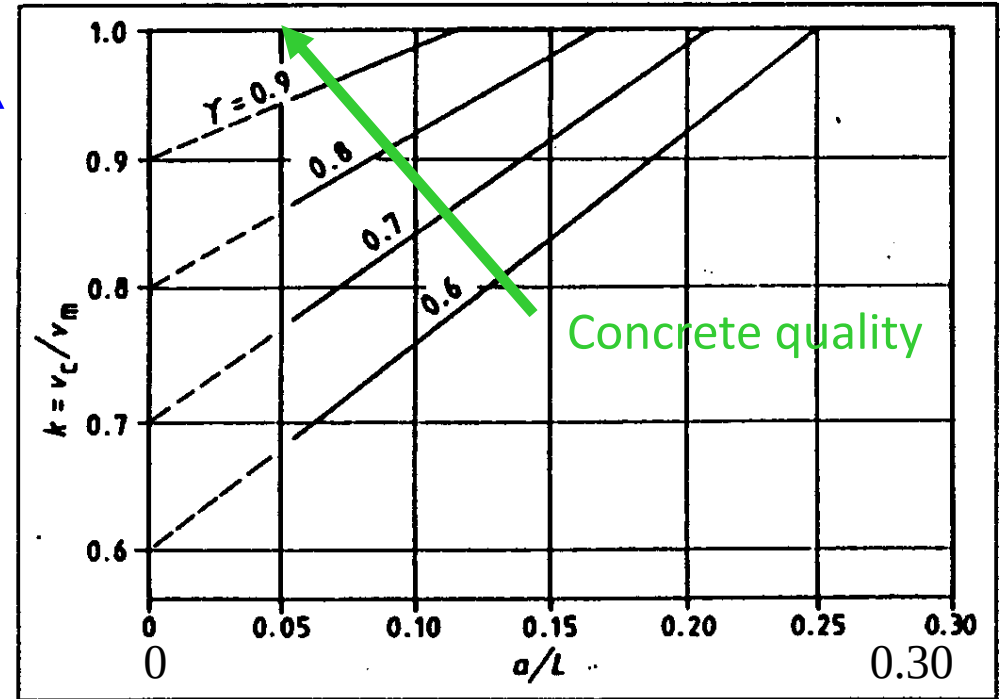


Influence of steel reinforcement on pulse velocity

1.0
bar parallel to pulse path



0.6
k



closer

a/L

Further away

Reinforcement

Influence by Temperature

Temperature °C	Correction to the measured pulse velocity	
	Air-dried concrete %	Water-saturated concrete %
60	+ 5	+ 4
40	+ 2	+ 1,7
10 to 30	0	0
0	− 0,5	− 1
− 4	− 1,5	− 7,5

Higher temperature, higher sound velocity
 between 0 and 30 Grad C only about +/- 1 %
 Below zero high influence